
Expression & purification of tagged ds-stapled peptides

Introduction

This protocols is for the periplasmic expression and purification of SUMO tagged disulphide stapled peptides using E. coli BL21, cloned using this [cloning protocol](#) for this [vector](#).

Materials

› Expression

- › Auto-induction media TBM-5052 (600 ml) filter sterilised
- › Kanamycin 50mg/ml
- › 6 x 96-deep wells plate (round bottom, sterile, 1.3 ml).

› Lysis

- › Sodium Deoxycholate Detergent (Sigma, Cat No. D6750)
- › 100 mM Tris-HCl
- › cComplete™ EDTA-free Protease Inhibitor Cocktail (Sigma-Aldrich, Cat No. 11836170001)
- › MiliQ water
- › 0.2 µm syringe filters

› Purification

- › 100 µl / well of NiNTA resin
- › IMAC buffers: Equilibration, Wash, Elute
- › 96 well 25 um fritted plate to hold HIS resin (this model from Agilent: Agilent Technologies 200953100 (same as for [cowboy biochem](#))
- › multi-well plate manifold [example](#)
- › AcropepAdv 2 ml 0.2 um WWPTFE 96-well filter plate (Ref 8782 - Pal Corporation)

Procedure

Expression

1. Prepare 650 mL TBM-5052 following the recipe in table 1 (extracted from the Baker Lab [wiki](#)), the recipes for 50xM and 50x5052, MgSO₄, trace metals can be found [here](#).

Table1. TBM-5052 recipe



	A	B
1	Reagent	Amount per L (ml)
2	TBII	1000.00
3	50×M salts	20.00
4	50×5052	20.00
5	1 M MgSO ₄	2.00
6	1000× Studier Trace Metals	0.20
7	100 mM thiamine	0.01
8	Kanamycin 50 mg/ml	2.00

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2. Fill 6 x 96 well plates with 1 ml of media per well and inoculate each replicate with 100 μ l of the overnight starter.
 3. Incubate overnight for 16 h at 22-25 $^{\circ}$ C, shaking on a plate shaker

Lysis

4. Prepare lysis buffer as specified in recipe on table 2, below

Table 2. Lysis buffer recipe



	A	B	C
1	DOC Lysis Buffer		
2	Reagent	Concentration	Quantity
3	Tris-HCL (mM)	100 mM	75 ml
4	Sodium deoxycholate (%w/v)	100	0.563 g
5	cOmplete Protease Inhibitor tablet	50x	1.5 tablet

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5. Centrifuge the plates at 2,000 x g for 5 min in a cool centrifuge at 8 -10 °C (limit the centrifugation to avoid cell wall/membrane disruption).
 6. Discard supernatant and resuspend the cell pellet in 400 ul of 100 mM Tris-HCl. Mix by swirling or slightly vortexing to avoid cell damage. Pool cells from 3 plates together to two plates with 1.2 ml of cell suspension.
 7. Pellet cells as in step 4 and remove as much liquid as possible by blotting on paper tissue for a few seconds
 8. Resuspend cell pellets with 350 µl lysis buffer and incubate for **4h** at room temperature on an orbital or lateral shaking platform at 25 rpm or low speed
 9. Centrifuge lysate at 4,000 x g for 25 min at 8°C
 10. Carefully remove the clarified lysate without disturbing the pellet and dispense in a single 96 well plate.

IMAC

11. Equilibrate the required amount of 50% Ni-NTA resin (10 ml per 96 well plate), by washing with 20 CV of milliQ water, and 20 CV of the equilibration buffer (see Table 3).

This can be either done by dispensing the resin on a 50 ml falcon and centrifuging at 500 x g for 2-3 min, remove supernatant with a pipette and add the new solution, swirl by mixing or you can dispense the resin in a Ni-NTA column and do the equilibration there by gravity flow or using vacuum.

Table 3. IMAC buffers

	A	B	C	D	E	F	G	H
1	Solutions	Stock Concentration (mM)	Equilibration buffer		Wash buffer		Elution buffer	
2			FC (mM)	Vol (mL)	FC (mM)	Vol (mL)	FC (mM)	Vol (mL)
3	Tris HCl	1000	20	10	25	12.5	25	1.25
4	NaCl	5000	250	25	300	30	250	2.5
5	Imidazole	2000	10	2.5	20	5	400	10
6	Final Vol (ml)			500		500		50

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12. Using a multichannel pipette Carefully pipette 100 μ l of 50% resin into each well with the lysate. Cover the plate and incubate on an orbital or lateral shaking platform for 40 min at 4°C.
 13. Dispense the lysate and resin onto a 25 μ m fritted plate placed on top of the vacuum manifold holder

*Make sure to mix well the lysate with the p1000 before dispensing to ensure the resin has not precipitated.
 14. Once in the fritted plate, open the vacuum to allow the lysate to flow through, for a few seconds, beware of not over-drying the resin.
 15. Wash 4-5 X with wash buffer.
 16. Dry the bottom of the fritted plate well by centrifuging a few seconds on top of an empty plate.
 17. Dispense 200 μ l of elution buffer onto each well, place the plate on top of a 400 μ L flat bottom plate and centrifuge at 500 x g for 2 min -- [Check]

PAUSE **If concentration is needed - do at this step**
 18. Dispense the elute onto a 0.2 μ m filter plate (Pall), placed on top of **conical bottom 96 well plate (not deep well - 96 well NUNC)** and centrifuge at 4,000 x g for 10 min.
 19. Run the filter elute through SEC using a 2 ml column on an AKTA (Fan Boy) or HPLC. Use SEC Jupyter notebooks to analyse data.

HPLC Setup

20. Have 1L of Filtered buffer and load it onto the Line A. Make sure that there is a minimum of 200 ml of seal and needle wash.
21. Label and place collection plates (384 deep-well) in the designated fraction collection drawers (FC1 or 2)
22. Place the IMAC elute plate on the drawer, noting the position (front or back) and drawer used (D1 -D4).
23. Open the acquisition programme :

On the instrument window

24. Hover on top of autosampler, right clic and clic on assign plates and empty any that is wrongly placed and look for the plate type where IMAC elute is (96 well NUNC if using the standard conical bottom), press ok and wait until it updates
25. Hover over the fraction collection tab and similarly, clic on reset fraction volumes, on the pop-up window, tick on the box of reset all and clic ok

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26. On the fraction collection tab, go to modify and on the pop-up window clic on each plate location to assign the plates used (384Agilent250uL) or leave the position empty - by assigning " No Plate" from the drop box to each of the position (control clic) and clic ok after correctly assigned.
27. CRITICAL **IMPORTANT!! Turn on Seal Wash →.** Go to quaternary pump - right clic and on the drop menu clic control and select seal wash to be set as periodic. **TURN OFF AFTER THE RUN**

On the sequence/methods windows

28. On the methods tab, clic the right icon on the top lefthand corner to open the sequence
29. Go to General - the 1st 3 folder on the menu are the default ones. Select the S75 column.
30. **EQUILIBRATE:** Clic - equilibrate from ETOH and double clic to OPEN. Check that the cycles are added to the run sequence and provide the path and file name for the run results.
31. Clic RUN and check the RUN tab to see if it is Acquiring already and the instrument is already ON
32. **SAMPLES RUN:** Again clic on Sequence tab folder icon and clic to open the s75 column and select s75_SAMPLES_96_MAKE_YOUR_OWN_COPY.sqx. Double clic to OPEN. Make sure that the VIAL is set to the correct drawer and position, for example if the samples are on the front of drawer 1, vial should be D1F. Make sure that all the wells are present up to H12. To make your own copy clic on the SAVE AS icon on the top left corner of the sequence window and save it on your own directory under a folder labelled METHODS. Also set the results folder path to your directory (under a RUNS folder) and give it a name e.g. date_samplespool. Press RUN.
33. **CLEAN UP:** Go to sequence tab, select s75 and select Clean 3C_water_NAOH_water_ETOH. Select the path of the method results to be SAVED on your equilibration folder. Clic RUN.
34. **SHUT DOWN:** On the bottom of the RUN window is the SHUT DOWN method, all is filed and only press SUBMIT SHUTDOWN.

Export data

35. Refer to: https://wiki.ipd.uw.edu/protocols/wet_lab/sec/hplc